

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

SECTION 1: Identification of the substance/mixture and of the company/undertaking

1.1 Product identifier

Trade name	: Triethylene Glycol Bottoms Column
Product code	: U120C
Synonyms	: Triethyleneglycol and higher homologues

1.2 Relevant identified uses of the substance or mixture and uses advised against

Use of the Sub- stance/Mixture	: Chemical intermediate.
Uses advised against	: This product must not be used in applications other than those listed in Section 1 without first seeking the advice of the supplier. This product must not be used in applications other than the above without first seeking the advice of the supplier.

1.3 Details of the supplier of the safety data sheet

Manufacturer/Supplier	: Shell Chemicals Europe B.V. PO Box 2334 3000 CH Rotterdam Netherlands
Telephone	: +31 (0)10 441 5137 / +31 (0)10 441 5191
Telefax	: +31 (0)20 716 8316 / +31 (0)20 713 9230
Contact for Safety Data Sheet	: sccmsds@shell.com

1.4 Emergency telephone number

+44 (0) 1235 239 670 (This telephone number is available 24 hours per day, 7 days per week)
Poisons Centre: 070 245 245

Other information	: This product is a Polymer which is exempt from the obligation to register under REACH in accordance with Article II, Section 9.
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SECTION 2: Hazards identification

2.1 Classification of the substance or mixture

Classification (REGULATION (EC) No 1272/2008)

Based on available data this substance / mixture does not meet the classification criteria.

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

2.2 Label elements

Labelling (REGULATION (EC) No 1272/2008)

Hazard pictograms : No Hazard Symbol required
Signal word : No signal word

Hazard statements : PHYSICAL HAZARDS:
Not classified as a physical hazard according to CLP criteria.
HEALTH HAZARDS:
Not classified as a health hazard under CLP criteria.
ENVIRONMENTAL HAZARDS:
Not classified as environmental hazard according to CLP criteria.

Precautionary statements : **Prevention:**
No precautionary phrases.
Response:
No precautionary phrases.
Storage:
No precautionary phrases.
Disposal:
No precautionary phrases.

2.3 Other hazards

Ecological information: The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Toxicological information: The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

Not classified as flammable but will burn.

SECTION 3: Composition/information on ingredients

3.1 Substances

Components

Chemical name	CAS-No. EC-No. Index-No. Registration number	Classification	Concentration (% w/w)
Triethylene glycol	112-27-6		32 - 50

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version 1.2 Revision Date.: 19.02.2026 SDS Number: 800001014446 Date of last issue: 06.03.2023
Print Date. 20.02.2026

	203-953-2 01-2119438366-35		
Tetraethylene glycol	112-60-7 203-989-9 01-2119475497-23		25 - 45
Polyethylene glycol	25322-68-3 500-038-2		0 - 43

SECTION 4: First aid measures

4.1 Description of first aid measures

- General advice : Not expected to be a health hazard when used under normal conditions.
- Protection of first-aiders : When administering first aid, ensure that you are wearing the appropriate personal protective equipment according to the incident, injury and surroundings.
- If inhaled : No treatment necessary under normal conditions of use.
If symptoms persist, obtain medical advice.
- In case of skin contact : Remove contaminated clothing. Flush exposed area with water and follow by washing with soap if available.
If persistent irritation occurs, obtain medical attention.
- In case of eye contact : Flush eye with copious quantities of water.
Remove contact lenses, if present and easy to do. Continue rinsing.
If persistent irritation occurs, obtain medical attention.
- If swallowed : In general no treatment is necessary unless large quantities are swallowed, however, get medical advice.

4.2 Most important symptoms and effects, both acute and delayed

- Symptoms : Not considered to be an inhalation hazard under normal conditions of use.
Possible respiratory irritation signs and symptoms may include a temporary burning sensation of the nose and throat, coughing, and/or difficulty breathing.
No specific hazards under normal use conditions.
Eye irritation signs and symptoms may include a burning sensation, redness, swelling, and/or blurred vision.

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

Skin irritation signs and symptoms may include a burning sensation, redness, or swelling.
Ingestion may result in nausea, vomiting and/or diarrhea.

4.3 Indication of any immediate medical attention and special treatment needed

Treatment : Call a doctor or poison control center for guidance.
Treat symptomatically.
May cause significant renal, respiratory, and CNS toxicity.
May cause significant acidosis.

SECTION 5: Firefighting measures

5.1 Extinguishing media

Suitable extinguishing media : Alcohol-resistant foam, water spray or fog. Dry chemical powder, carbon dioxide, sand or earth may be used for small fires only.

Unsuitable extinguishing media : Do not use water in a jet.

5.2 Special hazards arising from the substance or mixture

Specific hazards during fire-fighting : Material will not burn unless preheated.
Carbon monoxide may be evolved if incomplete combustion occurs.
Containers exposed to intense heat from fires should be cooled with large quantities of water.

5.3 Advice for firefighters

Special protective equipment for firefighters : Proper protective equipment including chemical resistant gloves are to be worn; chemical resistant suit is indicated if large contact with spilled product is expected. Self-Contained Breathing Apparatus must be worn when approaching a fire in a confined space. Select fire fighter's clothing approved to relevant Standards (e.g. Europe: EN469).

Specific extinguishing methods : Standard procedure for chemical fires.

Further information : Evacuate the area of all non-essential personnel.
Keep adjacent containers cool by spraying with water.

SECTION 6: Accidental release measures

6.1 Personal precautions, protective equipment and emergency procedures

Personal precautions : Observe all relevant local and international regulations.

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

Notify authorities if any exposure to the general public or the environment occurs or is likely to occur.

Local authorities should be advised if significant spillages cannot be contained.

6.1.1 For non emergency personnel:

Avoid contact with skin, eyes and clothing.

6.1.2 For emergency responders:

Avoid contact with skin, eyes and clothing.

6.2 Environmental precautions

Environmental precautions : Prevent from spreading or entering into drains, ditches or rivers by using sand, earth, or other appropriate barriers.
Use appropriate containment to avoid environmental contamination.
Ventilate contaminated area thoroughly.

6.3 Methods and material for containment and cleaning up

Methods for cleaning up : Contain run-off from residue flush and dispose of properly.
Soak up residue with an absorbent such as clay, sand or other suitable material.

For small liquid spills (< 1 drum), transfer by mechanical means to a labeled, sealable container for product recovery or safe disposal. Allow residues to evaporate or soak up with an appropriate absorbent material and dispose of safely. Remove contaminated soil and dispose of safely.

For large liquid spills (> 1 drum), transfer by mechanical means such as vacuum truck to a salvage tank for recovery or safe disposal. Do not flush away residues with water. Retain as contaminated waste. Allow residues to evaporate or soak up with an appropriate absorbent material and dispose of safely. Remove contaminated soil and dispose of safely.

6.4 Reference to other sections

For guidance on selection of personal protective equipment see Section 8 of this Safety Data Sheet.,
For guidance on disposal of spilled material see Section 13 of this Safety Data Sheet.

SECTION 7: Handling and storage

7.1 Precautions for safe handling

Technical measures : Avoid breathing of or direct contact with material. Only use in well ventilated areas. Wash thoroughly after handling. For guidance on selection of personal protective equipment see Section 8 of this Safety Data Sheet.
Use the information in this data sheet as input to a risk assessment of local circumstances to help determine appropri-

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

-
- | | | |
|-------------------------|---|---|
| Advice on safe handling | : | ate controls for safe handling, storage and disposal of this material.
Ensure that all local regulations regarding handling and storage facilities are followed.
Use local exhaust extraction over processing area.
Handle and open container with care in a well-ventilated area.
Do not empty into drains.
When handling product in drums, safety footwear should be worn and proper handling equipment should be used.
Handling Temperature:
Ambient. |
| Product Transfer | : | Keep containers closed when not in use. Do not pressurize drum containers to empty. |
| Hygiene measures | : | Wash hands before eating, drinking, smoking and using the toilet. Launder contaminated clothing before re-use. |

7.2 Conditions for safe storage, including any incompatibilities

- | | | |
|---|---|--|
| Requirements for storage areas and containers | : | Refer to section 15 for any additional specific legislation covering the packaging and storage of this product. |
| Further information on storage stability | : | Tanks must be clean, dry and rust-free.
Keep container tightly closed.
Must be stored in a diked (bunded) well-ventilated area, away from sunlight, ignition sources and other sources of heat.
Cleaning, inspection and maintenance of storage tanks is a specialist operation, which requires the implementation of strict procedures and precautions.
Drums should be stacked to a maximum of 3 high.
Storage Temperature:
Ambient. |
| Packaging material | : | Suitable material: Stainless steel., Mild steel., Carbon steel
Unsuitable material: Data not available |
| Container Advice | : | Containers, even those that have been emptied, can contain explosive vapours. Do not cut, drill, grind, weld or perform similar operations on or near containers. |

7.3 Specific end use(s)

- | | | |
|-----------------|---|-----------------|
| Specific use(s) | : | Not applicable. |
|-----------------|---|-----------------|

Ensure that all local regulations regarding handling and storage facilities are followed.

SECTION 8: Exposure controls/personal protection

8.1 Control parameters

Contains no substances with occupational exposure limit values.

Biological occupational exposure limits

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

No biological limit allocated.

8.2 Exposure controls

Engineering measures

The level of protection and types of controls necessary will vary depending upon potential exposure conditions. Select controls based on a risk assessment of local circumstances. Appropriate measures include:

Adequate ventilation to control airborne concentrations.

Where material is heated, sprayed or mist formed, there is greater potential for airborne concentrations to be generated.

Eye washes and showers for emergency use.

General Information

Always observe good personal hygiene measures, such as washing hands after handling the material and before eating, drinking, and/or smoking. Routinely wash work clothing and protective equipment to remove contaminants. Discard contaminated clothing and footwear that cannot be cleaned. Practice good housekeeping.

Define procedures for safe handling and maintenance of controls.

Educate and train workers in the hazards and control measures relevant to normal activities associated with this product.

Ensure appropriate selection, testing and maintenance of equipment used to control exposure, e.g. personal protective equipment, local exhaust ventilation.

Drain down system prior to equipment break-in or maintenance.

Retain drain downs in sealed storage pending disposal or subsequent recycle.

Personal protective equipment

Read in conjunction with the Exposure Scenario for your specific use contained in the Annex.

The provided information is made in consideration of the PPE regulation (Regulation (EU) 2016/425) and the CEN European Committee for Standardisation (CEN) standards.

Personal protective equipment (PPE) should meet recommended national standards. Check with PPE suppliers.

Eye protection : If material is handled such that it could be splashed into eyes, protective eyewear is recommended.
Approved to EU Standard EN166.

Hand protection

Remarks : Where hand contact with the product may occur the use of gloves approved to relevant standards (e.g. Europe: EN374, US: F739) made from the following materials may provide suitable chemical protection. Longer term protection: Nitrile rubber gloves. Incidental contact/Splash protection: PVC or neoprene rubber gloves. For continuous contact we recommend gloves with breakthrough time of more than 240 minutes with preference for > 480 minutes where suitable gloves can be identified. For short-term/splash protection we recommend the same but recognize that suitable gloves offering this level of protection may not be available and in this

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

		case a lower breakthrough time maybe acceptable so long as appropriate maintenance and replacement regimes are followed. Glove thickness is not a good predictor of glove resistance to a chemical as it is dependent on the exact composition of the glove material. Glove thickness should be typically greater than 0.35 mm depending on the glove make and model. Suitability and durability of a glove is dependent on usage, e.g. frequency and duration of contact, chemical resistance of glove material, dexterity. Always seek advice from glove suppliers. Contaminated gloves should be replaced. Personal hygiene is a key element of effective hand care. Gloves must only be worn on clean hands. After using gloves, hands should be washed and dried thoroughly. Application of a non-perfumed moisturizer is recommended.
Skin and body protection	:	Skin protection is not ordinarily required beyond standard work clothes. It is good practice to wear chemical resistant gloves. Protective clothing approved to EU Standard EN14605.
Respiratory protection	:	If engineering controls do not maintain airborne concentrations to a level which is adequate to protect worker health, select respiratory protection equipment suitable for the specific conditions of use and meeting relevant legislation. Check with respiratory protective equipment suppliers. Where air-filtering respirators are unsuitable (e.g. airborne concentrations are high, risk of oxygen deficiency, confined space) use appropriate positive pressure breathing apparatus. Where air-filtering respirators are suitable, select an appropriate combination of mask and filter. If air-filtering respirators are suitable for conditions of use: Select a filter suitable for the combination of organic gases and vapours and particles meeting EN14387 and EN143 [Filter type A/P for use against certain organic gases and vapours with a boiling point >65°C (149°F) and for use against particles].
Thermal hazards	:	Not applicable

SECTION 9: Physical and chemical properties

9.1 Information on basic physical and chemical properties

Physical state	:	Slightly viscous liquid.
Colour	:	brown, dark brown
Odour	:	mild
Odour Threshold	:	Data not available
Melting point/ range	:	Data not available
Boiling point/boiling range	:	294 - 296 °C

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

Flammability

Flammability (solid, gas) : Not classified as flammable but will burn.

Lower explosion limit and upper explosion limit / flammability limit

Upper explosion limit /
Upper flammability limit : Data not available

Lower explosion limit /
Lower flammability limit : Data not available

Flash point : > 166 °C

Auto-ignition temperature : 323 °C

Decomposition temperature
Decomposition temperature : Data not available

pH : Not applicable

Viscosity

Viscosity, dynamic : Data not available

Viscosity, kinematic : Data not available

Solubility(ies)

Water solubility : completely miscible

Partition coefficient: n-
octanol/water : Data not available

Vapour pressure : < 1 Pa (20 °C)

Relative density : 1,13 - 1,14
Method: ASTM D4052

Density : Data not available

Relative vapour density : Data not available

Particle characteristics

Particle size : Data not available

9.2 Other information

Explosive properties : Not applicable

Oxidizing properties : Data not available

Evaporation rate : Data not available

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

Conductivity : Electrical conductivity: > 10,000 pS/m

A number of factors, for example liquid temperature, presence of contaminants, and anti-static additives can greatly influence the conductivity of a liquid, This material is not expected to be a static accumulator.

Surface tension : Data not available

Molecular weight : Data not available

SECTION 10: Stability and reactivity

10.1 Reactivity

The product does not pose any further reactivity hazards in addition to those listed in the following sub-paragraph.

10.2 Chemical stability

No hazardous reaction is expected when handled and stored according to provisions
Oxidises on contact with air.

10.3 Possibility of hazardous reactions

Hazardous reactions : None known.

10.4 Conditions to avoid

Conditions to avoid : Extremes of temperature and direct sunlight.

Product cannot ignite due to static electricity.

10.5 Incompatible materials

Materials to avoid : Strong oxidising agents.
Strong acids.
Strong bases.

10.6 Hazardous decomposition products

Thermal decomposition is highly dependent on conditions. A complex mixture of airborne solids, liquids and gases including carbon monoxide, carbon dioxide, sulphur oxides and unidentified organic compounds will be evolved when this material undergoes combustion or thermal or oxidative degradation.

SECTION 11: Toxicological information

11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008

Information on likely routes of exposure : Exposure may occur via inhalation, ingestion, skin absorption, skin or eye contact, and accidental ingestion.

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

Acute toxicity

Product:

- | | |
|---------------------------|---|
| Acute oral toxicity | : LD 50 (Rat, male and female): > 2.000 mg/kg
Method: Literature data
Remarks: Based on available data, the classification criteria are not met. |
| Acute inhalation toxicity | : LC 50 (Rat, male and female): > 5 mg/l
Exposure time: 4 h
Test atmosphere: Aerosol
Method: Acceptable non-standard method.
Remarks: Based on available data, the classification criteria are not met. |
| Acute dermal toxicity | : LD 50 (Rabbit, male and female): 16 ml/kg bw
Method: Acceptable non-standard method.
Remarks: Based on available data, the classification criteria are not met. |

Components:

Triethylene glycol:

- | | |
|---------------------------|---|
| Acute oral toxicity | : LD 50 (Rat, male and female): > 2.000 mg/kg
Method: Literature data
Remarks: Based on available data, the classification criteria are not met. |
| Acute inhalation toxicity | : LC 50 (Rat, male and female): > 5 mg/l
Exposure time: 4 h
Test atmosphere: Aerosol
Method: Acceptable non-standard method.
Remarks: Based on available data, the classification criteria are not met. |
| Acute dermal toxicity | : LD 50 (Rabbit, male and female): 16 ml/kg bw
Method: Acceptable non-standard method.
Remarks: Based on available data, the classification criteria are not met. |

Skin corrosion/irritation

Product:

- | | |
|---------|---|
| Species | : Rabbit |
| Method | : Literature data |
| Remarks | : Slightly irritating.
Insufficient to classify. |

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

Components:

Triethylene glycol:

Species	:	Rabbit
Method	:	Literature data
Remarks	:	Slightly irritating. Insufficient to classify.

Serious eye damage/eye irritation

Product:

Species	:	Rabbit
Method	:	Literature data
Remarks	:	Slightly irritating. Insufficient to classify.

Components:

Triethylene glycol:

Species	:	Rabbit
Method	:	Literature data
Remarks	:	Slightly irritating. Insufficient to classify.

Respiratory or skin sensitisation

Product:

Species	:	Guinea pig
Method	:	Test(s) equivalent or similar to OECD Test Guideline 406
Remarks	:	Based on available data, the classification criteria are not met.

Components:

Triethylene glycol:

Species	:	Guinea pig
Method	:	Test(s) equivalent or similar to OECD Test Guideline 406
Remarks	:	Based on available data, the classification criteria are not met.

Germ cell mutagenicity

Product:

Genotoxicity in vitro	:	Method: OECD Test Guideline 471 Remarks: Based on available data, the classification criteria are not met.
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SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

Method: Test(s) equivalent or similar to OECD Test Guideline 473

Remarks: Based on available data, the classification criteria are not met.

Method: Test(s) equivalent or similar to OECD Test Guideline 479

Remarks: Based on available data, the classification criteria are not met.

Germ cell mutagenicity- Assessment : This product does not meet the criteria for classification in categories 1A/1B.

Components:

Triethylene glycol:

Genotoxicity in vitro : Method: OECD Test Guideline 471
Remarks: Based on available data, the classification criteria are not met.

Method: Test(s) equivalent or similar to OECD Test Guideline 473

Remarks: Based on available data, the classification criteria are not met.

Method: Test(s) equivalent or similar to OECD Test Guideline 479

Remarks: Based on available data, the classification criteria are not met.

Germ cell mutagenicity- Assessment : This product does not meet the criteria for classification in categories 1A/1B.

Carcinogenicity

Product:

Species	:	Rat, male and female
Application Route	:	Oral
Method	:	Literature data
Test substance	:	Diethylene glycol
Remarks	:	Based on available data, the classification criteria are not met.

Carcinogenicity - Assessment : This product does not meet the criteria for classification in categories 1A/1B.

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version 1.2 Revision Date.: 19.02.2026 SDS Number: 800001014446 Date of last issue: 06.03.2023
Print Date. 20.02.2026

Components:

Triethylene glycol:

Species : Rat, male and female
Application Route : Oral
Method : Literature data
Test substance : Diethylene glycol
Remarks : Based on available data, the classification criteria are not met.

Carcinogenicity - Assessment : This product does not meet the criteria for classification in categories 1A/1B.

Material	GHS/CLP Carcinogenicity Classification
Triethylene glycol	No carcinogenicity classification.
Tetraethylene glycol	No carcinogenicity classification.
Polyethylene glycol	No carcinogenicity classification.

Reproductive toxicity

Product:

Effects on fertility : Species: Mouse
Sex: male and female
Application Route: Oral

Method: Acceptable non-standard method.
Remarks: Based on available data, the classification criteria are not met.

Reproductive toxicity - Assessment : This product does not meet the criteria for classification in categories 1A/1B.

Components:

Triethylene glycol:

Effects on fertility : Species: Mouse
Sex: male and female
Application Route: Oral

Method: Acceptable non-standard method.
Remarks: Based on available data, the classification criteria are not met.

Reproductive toxicity - Assessment : This product does not meet the criteria for classification in categories 1A/1B.

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

STOT - single exposure

Product:

Remarks : Based on available data, the classification criteria are not met.
Inhalation of vapours or mists may cause irritation to the respiratory system.

Components:

Triethylene glycol:

Remarks : Based on available data, the classification criteria are not met.
Inhalation of vapours or mists may cause irritation to the respiratory system.

STOT - repeated exposure

Product:

Remarks : Based on available data, the classification criteria are not met.

Components:

Triethylene glycol:

Remarks : Based on available data, the classification criteria are not met.

Repeated dose toxicity

Product:

Species : Rat, male and female
Application Route : Oral
Method : Test(s) equivalent or similar to OECD Test Guideline 408
Target Organs : No specific target organs noted

Species : Rat, male and female
Application Route : Inhalation
Test atmosphere : Aerosol
Method : Acceptable non-standard method.
Test substance : PEG 200
Target Organs : No specific target organs noted

Components:

Triethylene glycol:

Species : Rat, male and female
Application Route : Oral
Method : Test(s) equivalent or similar to OECD Test Guideline 408

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

Target Organs	:	No specific target organs noted
Species	:	Rat, male and female
Application Route	:	Inhalation
Test atmosphere	:	Aerosol
Method	:	Acceptable non-standard method.
Test substance	:	PEG 200
Target Organs	:	No specific target organs noted

Aspiration toxicity

Product:

Based on available data, the classification criteria are not met.

Components:

Triethylene glycol:

Based on available data, the classification criteria are not met.

11.2 Information on other hazards

Endocrine disrupting properties

Product:

Assessment	:	The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.
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Further information

Product:

Remarks	:	Classifications by other authorities under varying regulatory frameworks may exist.
Remarks	:	Unless indicated otherwise, the data presented is representative of the product as a whole, rather than for individual component(s).

Components:

Triethylene glycol:

Remarks	:	Classifications by other authorities under varying regulatory frameworks may exist.
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SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

SECTION 12: Ecological information

12.1 Toxicity

Product:

- | | | |
|--|---|--|
| Toxicity to fish | : | LC50 (Lepomis macrochirus (Bluegill sunfish)): > 10.000 mg/l
Exposure time: 96 h
Method: Test(s) equivalent or similar to OECD Guideline 203
Remarks: Practically non toxic:
LL/EL/IL50 > 100 mg/l |
| Toxicity to daphnia and other aquatic invertebrates | : | (Daphnia magna (Water flea)): > 10.000 mg/l
Exposure time: 48 h
Method: Other guideline method.
Remarks: Practically non toxic:
LL/EL/IL50 > 100 mg/l |
| Toxicity to algae/aquatic plants | : | EC50 (Selenastrum capricornutum (green algae)): 6.500 - 13.000 mg/l
Exposure time: 96 h
Method: Other guideline method.
Remarks: Practically non toxic:
LL/EL/IL50 > 100 mg/l |
| Toxicity to fish (Chronic toxicity) | : | NOEC: 15.380 mg/l
Exposure time: 7 d
Species: Pimephales promelas (fathead minnow)
Method: Other guideline method.
Remarks: NOEC/NOEL/EL10 > 100 mg/l |
| Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) | : | NOEC: > 15.000 mg/l
Exposure time: 21 d
Species: Daphnia magna (Water flea)
Method: Other guideline method.
Remarks: NOEC/NOEL/EL10 > 100 mg/l |
| Toxicity to microorganisms | : | EC10 (Activated sludge): > 1.995 mg/l
Exposure time: 0,5 h
Method: Other guideline method.
Remarks: Practically non toxic:
LL/EL/IL50 > 100 mg/l |

Components:

Triethylene glycol:

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

Toxicity to fish	: LC50 (Lepomis macrochirus (Bluegill sunfish)): > 10.000 mg/l Exposure time: 96 h Method: Test(s) equivalent or similar to OECD Guideline 203 Remarks: Practically non toxic: LL/EL/IL50 > 100 mg/l
Toxicity to daphnia and other aquatic invertebrates	: (Daphnia magna (Water flea)): > 10.000 mg/l Exposure time: 48 h Method: Other guideline method. Remarks: Practically non toxic: LL/EL/IL50 > 100 mg/l
Toxicity to algae/aquatic plants	: EC50 (Selenastrum capricornutum (green algae)): 6.500 - 13.000 mg/l Exposure time: 96 h Method: Other guideline method. Remarks: Practically non toxic: LL/EL/IL50 > 100 mg/l
Toxicity to microorganisms	: EC10 (Activated sludge): > 1.995 mg/l Exposure time: 0,5 h Method: Other guideline method. Remarks: Practically non toxic: LL/EL/IL50 > 100 mg/l
Toxicity to fish (Chronic toxicity)	: NOEC: 15.380 mg/l Exposure time: 7 d Species: Pimephales promelas (fathead minnow) Method: Other guideline method. Remarks: NOEC/NOEL/EL10 > 100 mg/l
Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity)	: NOEC: > 15.000 mg/l Exposure time: 21 d Species: Daphnia magna (Water flea) Method: Other guideline method. Remarks: NOEC/NOEL/EL10 > 100 mg/l

12.2 Persistence and degradability

Product:

Biodegradability	: Biodegradation: 90 - 100 % Exposure time: 10 d Method: OECD Test Guideline 301A Remarks: Readily biodegradable. Oxidises rapidly by photo-chemical reactions in air.
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Components:

Triethylene glycol:

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

Biodegradability : Biodegradation: 90 - 100 %
Exposure time: 10 d
Method: OECD Test Guideline 301A
Remarks: Readily biodegradable.
Oxidises rapidly by photo-chemical reactions in air.

12.3 Bioaccumulative potential

Product:

Bioaccumulation : Remarks: Does not have the potential to bioaccumulate significantly.

Components:

Triethylene glycol:

Bioaccumulation : Remarks: Does not have the potential to bioaccumulate significantly.

12.4 Mobility in soil

Product:

Mobility : Remarks: If product enters soil, it will be highly mobile and may contaminate groundwater., Sinks in water.

Components:

Triethylene glycol:

Mobility : Remarks: If product enters soil, it will be highly mobile and may contaminate groundwater., Sinks in water.

12.5 Results of PBT and vPvB assessment

Not relevant

12.6 Endocrine disrupting properties

Product:

Assessment : The substance/mixture does not contain components considered to have endocrine disrupting properties according to REACH Article 57(f) or Commission Delegated regulation (EU) 2017/2100 or Commission Regulation (EU) 2018/605 at levels of 0.1% or higher.

12.7 Other adverse effects

Product:

Additional ecological information : Unless indicated otherwise, the data presented is representative of the product as a whole, rather than for individual component(s).

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

SECTION 13: Disposal considerations

13.1 Waste treatment methods

- | | |
|------------------------|---|
| Product | : Recover or recycle if possible.
It is the responsibility of the waste generator to determine the toxicity and physical properties of the material generated to determine the proper waste classification and disposal methods in compliance with applicable regulations.
Remove all packaging for recovery or waste disposal.
Waste product should not be allowed to contaminate soil or ground water, or be disposed of into the environment.
Do not dispose of tank water bottoms by allowing them to drain into the ground. This will result in soil and groundwater contamination.
Do not dispose into the environment, in drains or in water courses.
Waste arising from a spillage or tank cleaning should be disposed of in accordance with prevailing regulations, preferably to a recognised collector or contractor. The competence of the collector or contractor should be established beforehand.

Disposal should be in accordance with applicable regional, national, and local laws and regulations.
Local regulations may be more stringent than regional or national requirements and must be complied with.

MARPOL - see International Convention for the Prevention of Pollution from Ships (MARPOL 73/78) which provides technical aspects at controlling pollutions from ships. |
| Contaminated packaging | : Dispose in accordance with prevailing regulations, preferably to a recognized collector or contractor. The competence of the collector or contractor should be established beforehand. |

SECTION 14: Transport information

14.1 UN number or ID number

- | | |
|------|-------------------------------------|
| ADN | : Not regulated as a dangerous good |
| ADR | : Not regulated as a dangerous good |
| RID | : Not regulated as a dangerous good |
| IMDG | : Not regulated as a dangerous good |

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

IATA	: Not regulated as a dangerous good
14.2 UN proper shipping name	
ADN	: Not regulated as a dangerous good
ADR	: Not regulated as a dangerous good
RID	: Not regulated as a dangerous good
IMDG	: Not regulated as a dangerous good
IATA	: Not regulated as a dangerous good
14.3 Transport hazard class(es)	
ADN	: Not regulated as a dangerous good
ADR	: Not regulated as a dangerous good
RID	: Not regulated as a dangerous good
IMDG	: Not regulated as a dangerous good
IATA	: Not regulated as a dangerous good
14.4 Packing group	
ADN	: Not regulated as a dangerous good
CDNI Inland Water Waste Agreement	: NST 8963 Glycols unspecified
ADR	: Not regulated as a dangerous good
RID	: Not regulated as a dangerous good
IMDG	: Not regulated as a dangerous good
IATA	: Not regulated as a dangerous good
14.5 Environmental hazards	
ADN	: Not regulated as a dangerous good
ADR	: Not regulated as a dangerous good
RID	: Not regulated as a dangerous good
IMDG	: Not regulated as a dangerous good
14.6 Special precautions for user	
Remarks	: Special Precautions: Refer to Section 7, Handling & Storage, for special precautions which a user needs to be aware of or needs to comply with in connection with transport.
14.7 Maritime transport in bulk according to IMO instruments	
Pollution category	: Not applicable
Ship type	: Not applicable
Product name	: Not applicable
Additional Information	: This product may be transported under nitrogen blanketing. Nitrogen is an odourless and invisible gas. Exposure to nitrogen enriched atmospheres displaces available oxygen which

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

may cause asphyxiation or death. Personnel must observe strict safety precautions when involved with a confined space entry.

SECTION 15: Regulatory information

15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

Other regulations:

The regulatory information is not intended to be comprehensive. Other regulations may apply to this material.

The components of this product are reported in the following inventories:

AU AIIC	: Listed
CA. DSL	: Listed
CN IECSC	: Listed
JP ENCS	: Listed
NZ NZIoC	: Listed
PH PICCS	: Listed
US TSCA	: Listed

15.2 Chemical safety assessment.

A Chemical Safety Assessment is not required for this substance.

SECTION 16: Other information

Full text of other abbreviations

ADN - European Agreement concerning the International Carriage of Dangerous Goods by Inland Waterways; ADR - Agreement concerning the International Carriage of Dangerous Goods by Road; AIIC - Australian Inventory of Industrial Chemicals; ASTM - American Society for the Testing of Materials; bw - Body weight; CLP - Classification Labelling Packaging Regulation; Regulation (EC) No 1272/2008; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECHA - European Chemicals Agency; EC-Number - European Community number; ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergen-

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

cy Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; GHS - Globally Harmonised System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organisation; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardisation; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NZIoC - New Zealand Inventory of Chemicals; OECD - Organisation for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; RID - Regulations concerning the International Carriage of Dangerous Goods by Rail; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; SVHC - Substance of Very High Concern; TCSI - Taiwan Chemical Substance Inventory; TECI - Thailand Existing Chemicals Inventory; TRGS - Technical Rule for Hazardous Substances; TSCA - Toxic Substances Control Act (United States); UN - United Nations; vPvB - Very Persistent and Very Bioaccumulative

Further information

- Training advice : Provide adequate information, instruction and training for operators.
- Other information : For Industry guidance and tools on REACH please visit the CEFIC website at <http://cefic.org/Industry-support>.
The substance does not fulfill all screening criteria for persistence, bioaccumulation and toxicity and hence is not considered to be PBT or vPvB.
A vertical bar (|) in the left margin indicates an amendment from the previous version.
- Sources of key data used to compile the Safety Data Sheet : The quoted data are from, but not limited to, one or more sources of information (e.g. toxicological data from Shell Health Services, material suppliers' data, CONCAWE, EU IUCLID data base, EC 1272 regulation, etc).

Identified Uses according to the Use Descriptor System

Uses - Worker

- Title : Manufacture of substance
- Industrial

Uses - Worker

- Title : Use as an intermediate

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version	Revision Date.:	SDS Number:	Date of last issue: 06.03.2023
1.2	19.02.2026	800001014446	Print Date. 20.02.2026

- Industrial

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SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version 1.2 Revision Date.: 19.02.2026 SDS Number: 800001014446 Date of last issue: 06.03.2023
Print Date. 20.02.2026

Exposure Scenario - Worker

300000000979	
SECTION 1	EXPOSURE SCENARIO TITLE
Title	Manufacture of substance- Industrial
Use Descriptor	Sector of Use: SU 3, SU8, SU9 Process Categories: PROC 1, PROC 2, PROC 3, PROC 4, PROC 8a, PROC 8b, PROC 15 Environmental Release Categories: ERC1, ERC4
Scope of process	Manufacture of the substance or use as a process chemical or extraction agent. Includes recycling/ recovery, material transfers, storage, maintenance and loading (including marine vessel/barge, road/rail car and bulk container), sampling and associated laboratory activities.

SECTION 2	OPERATIONAL CONDITIONS AND RISK MANAGEMENT MEASURES	
Section 2.1	Control of Worker Exposure	
Product Characteristics		
Physical form of product	Liquid, vapour pressure < 0.5 kPa at STP	
Concentration of the Substance in Mixture/Article	Covers use of substance/product up to 100% (unless stated differently).,	
Frequency and Duration of Use		
Covers daily exposures up to 8 hours (unless stated differently).		
Other Operational Conditions affecting Exposure		
Assumes a good basic standard of occupational hygiene is implemented. Assumes use at not more than 20°C above ambient temperature (unless stated differently).		
Contributing Scenarios	Risk Management Measures	
General measures (eye irritants).	Use suitable eye protection.	
General measures (skin irritants).	Avoid direct skin contact with product. Identify potential areas for indirect skin contact. Wear gloves (tested to EN374) if hand contact with substance likely. Clean up contamination/spills as soon as they occur. Wash off any skin contamination immediately. Provide basic employee training to prevent / minimise exposures and to report any skin problems that may develop.	
General exposures (closed systems)elevated temperatureContinuous process	No other specific measures identified.	
General exposures (closed systems)General measures	Ensure that no inhalable aerosols are generated.	

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version 1.2 Revision Date.: 19.02.2026 SDS Number: 800001014446 Date of last issue: 06.03.2023
Print Date. 20.02.2026

(skin irritants).elevated temperatureContinuous process	
General exposures.Use in contained batch processes	No other specific measures identified.
General exposures.Batch process	Ensure that no inhalable aerosols are generated.
Process samplingelevated temperature	Use a sampling system designed to control exposure.
Equipment cleaning and maintenance	Retain drain downs in sealed storage pending disposal or for subsequent recycle.
Bulk transfersDedicated facility	Transfer via enclosed lines. Clear transfer lines prior to de-coupling.
Bulk product storage(closed systems)	No other specific measures identified.
Laboratory activities	No other specific measures identified.
Section 2.2	Control of Environmental Exposure
No exposure assessment presented for the environment.	

SECTION 3	EXPOSURE ESTIMATION
Section 3.1 - Health	
The ECETOC TRA tool has been used to estimate workplace exposures unless otherwise indicated.	

Section 3.2 -Environment
No exposure assessment presented for the environment.

SECTION 4	GUIDANCE TO CHECK COMPLIANCE WITH THE EXPOSURE SCENARIO
Section 4.1 - Health	
Predicted exposures are not expected to exceed the DN(M)EL when the Risk Management Measures/Operational Conditions outlined in Section 2 are implemented. Where other Risk Management Measures/Operational Conditions are adopted, then users should ensure that risks are managed to at least equivalent levels.	

Section 4.2 -Environment
No exposure assessment presented for the environment.

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version 1.2 Revision Date.: 19.02.2026 SDS Number: 800001014446 Date of last issue: 06.03.2023
Print Date. 20.02.2026

Exposure Scenario - Worker

300000000980	
SECTION 1	EXPOSURE SCENARIO TITLE
Title	Use as an intermediate- Industrial
Use Descriptor	Sector of Use: SU 3, SU8, SU9 Process Categories: PROC 1, PROC 2, PROC 3, PROC 4, PROC 5, PROC 8a, PROC 8b, PROC 9, PROC 15 Environmental Release Categories: ERC6a
Scope of process	Use of substance as an intermediate (not related to Strictly Controlled Conditions). Includes recycling/ recovery, material transfers, storage, sampling, associated laboratory activities, maintenance and loading (including marine vessel/barge, road/rail car and bulk container).

SECTION 2	OPERATIONAL CONDITIONS AND RISK MANAGEMENT MEASURES	
Section 2.1	Control of Worker Exposure	
Product Characteristics		
Physical form of product	Liquid, vapour pressure < 0.5 kPa at STP	
Concentration of the Substance in Mixture/Article	Covers use of substance/product up to 100% (unless stated differently).,	
Frequency and Duration of Use		
Covers daily exposures up to 8 hours (unless stated differently).		
Other Operational Conditions affecting Exposure		
Assumes a good basic standard of occupational hygiene is implemented. Assumes use at not more than 20°C above ambient temperature (unless stated differently).		
Contributing Scenarios	Risk Management Measures	
General measures (eye irritants).	Use suitable eye protection.	
General measures (skin irritants).	Avoid direct skin contact with product. Identify potential areas for indirect skin contact. Wear gloves (tested to EN374) if hand contact with substance likely. Clean up contamination/spills as soon as they occur. Wash off any skin contamination immediately. Provide basic employee training to prevent / minimise exposures and to report any skin problems that may develop.	
Bulk transfersDedicated facility	Transfer via enclosed lines. Clear transfer lines prior to de-coupling.	
Bulk product storage(closed systems)	No other specific measures identified.	

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version 1.2 Revision Date.: 19.02.2026 SDS Number: 800001014446 Date of last issue: 06.03.2023
Print Date. 20.02.2026

Product sampling.	Use a sampling system designed to control exposure.
Laboratory activities	No other specific measures identified.
Mixing operations (open systems)	Ensure that no inhalable aerosols are generated.
General exposures (closed systems)Continuous process	No other specific measures identified.
General exposures (closed systems)General measures (skin irritants).Continuous process	Ensure that no inhalable aerosols are generated.
General exposures.Use in contained batch processes	No other specific measures identified.
General exposures.Batch process	Ensure that no inhalable aerosols are generated.
Equipment cleaning and maintenance	Retain drain downs in sealed storage pending disposal or for subsequent recycle.
Section 2.2	Control of Environmental Exposure
No exposure assessment presented for the environment.	

SECTION 3	EXPOSURE ESTIMATION
Section 3.1 - Health	
The ECETOC TRA tool has been used to estimate workplace exposures unless otherwise indicated.	

Section 3.2 -Environment	
No exposure assessment presented for the environment.	

SECTION 4	GUIDANCE TO CHECK COMPLIANCE WITH THE EXPOSURE SCENARIO
Section 4.1 - Health	
Predicted exposures are not expected to exceed the DN(M)EL when the Risk Management Measures/Operational Conditions outlined in Section 2 are implemented. Where other Risk Management Measures/Operational Conditions are adopted, then users should ensure that risks are managed to at least equivalent levels.	

Section 4.2 -Environment	
No exposure assessment presented for the environment.	

SAFETY DATA SHEET.

According to EC No 1907/2006 as amended as at the date of this SDS

Triethylene Glycol Bottoms Column

Version
1.2

Revision Date.:
19.02.2026

SDS Number:
800001014446

Date of last issue: 06.03.2023
Print Date. 20.02.2026
